

Module – 4

Metaphors and Prototyping

- Metaphor method: Theory and methodology of concept generation,
- Blend method & Thematic Method.
- Articulating design ideas: Storytelling;
- Introduction to Sketching & Dynamic Diagrams;
- K Scripts,
- Introduction to Prototyping & Visualization Design Tools



'For me design is a question mark.
That is how I start every new project.'

Gijs Bakker, 2007
(Dutch jewellery designer)





Few products designed by Mr. Bakker



Metaphor Method in Concept generation

A metaphor is a rhetorical figure representing one concept by way of another concept, which is considered similar to the concept to be represented. *Metaphors are often used in design.*

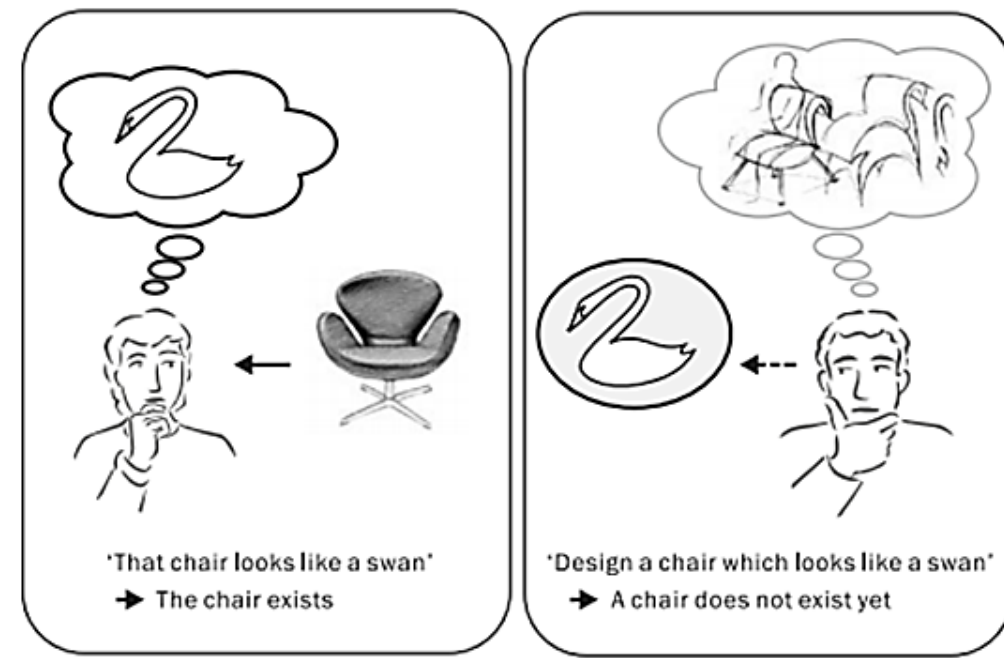
For example, in the figure shows a famous chair called the Swan Chair. This chair was created by the Danish designer Arne Jacobsen for a hotel, and it remains very popular. As its name conveys, the Swan Chair is a chair which approximates the shape of a swan in order to convey the quality of a swan (i.e. elegance) and can be represented by the framework for a metaphor, such as ‘**the chair is a swan**’



Types of Metaphors

If a person says **“that chair looks like a swan”**

when he or she looks at the chair, it refers to an existing chair in the picture. On the other hand, if a person says this while he or she is designing the chair, then the chair is a concept which does not yet exist.



In this situation, **‘the new chair which is to be designed is a swan’** is a more suitable representation. In such a case, the similarity (implicit common features; e.g. elegant shape) between the chair and a swan play an important role. On the basis of the above considerations, we can say that, in the design process using metaphors, a new concept will be generated which will be similar to the concept to be assimilated to. A new concept of a chair is generated (referring to a swan or imitating a swan) on the basis of the similarity between the chair to be designed and a swan.

“First order Concept generation”

Higher Order Concept Generation



A first-order concept generation is useful only to create a subspecies of an existing object, since it cannot extend beyond the category of the existing product.

Abstract concept generated by combining multiple abstract concepts is called a high-order abstract concept, and the high-order concept generation is the process of generating a new concept on the basis of this high-order abstract concept.



Difference between “White Tomato” and “Powdered Ketchup”

A high-order concept generation is understood to have its basis on the dissimilarity-recognition process.

Blending and Thematic Method



High-order abstract concepts can be interpreted as abstract concepts **involving an innovative concept** which inherits partial properties from both the two base concepts but is different from the two base concepts. *We define the concept generation based on this process as concept blending or blending method.*

The concept of **“powdered ketchup”** is an example of concept blending. This new concept is understood to be generated by blending the different properties of snow (**abstract concept: snowflake**) and tomato (**abstract concept: flavouring**).

On the other hand, according to the latter relation between the two base concepts, high-order abstract concepts can be interpreted as abstract concepts involving an innovative concept which is generated from the thematic scenes (situations, roles, etc.) of the base concepts. We define the concept generation based on the thematic relation as concept integration in thematic relation.

WHICH ONE ARE YOU ?

**LEFT
BRAIN**



**RIGHT
BRAIN**



Comparison of Left-Brained and Right-Brained Thinking

Critical Thinking (Left Brain)

Logical, analytic, judgmental process

Linear

Leads to only one solution

Considers only relevant information

Movement is made in a sequential, rule-based manner

Embodies scientific principles

Classifications and labels are rigid

Vertical

Convergent

Creative Thinking (Right Brain)

Generative, suspended judgment

Associative

Creates many possible solutions

Considers broad range of information

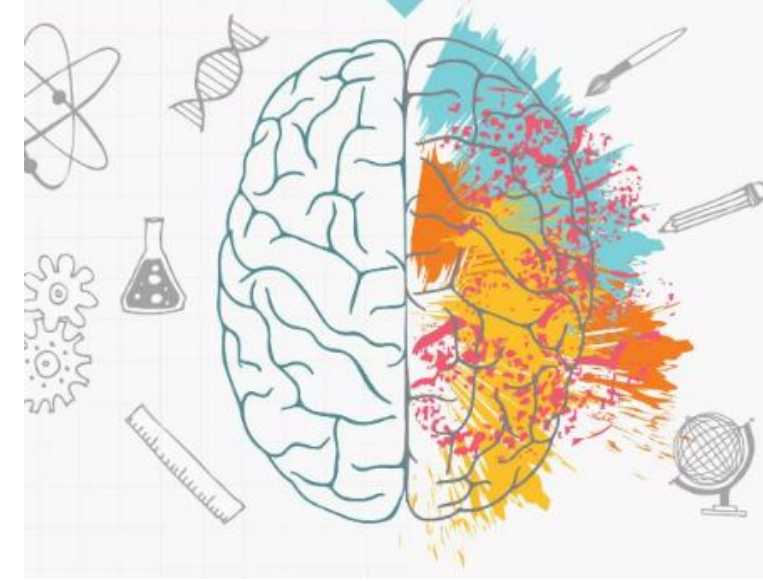
Movement is made in a more random pattern

Heavily influenced by symbols and imagery

Reclassifies objects to generate ideas

Lateral

Divergent

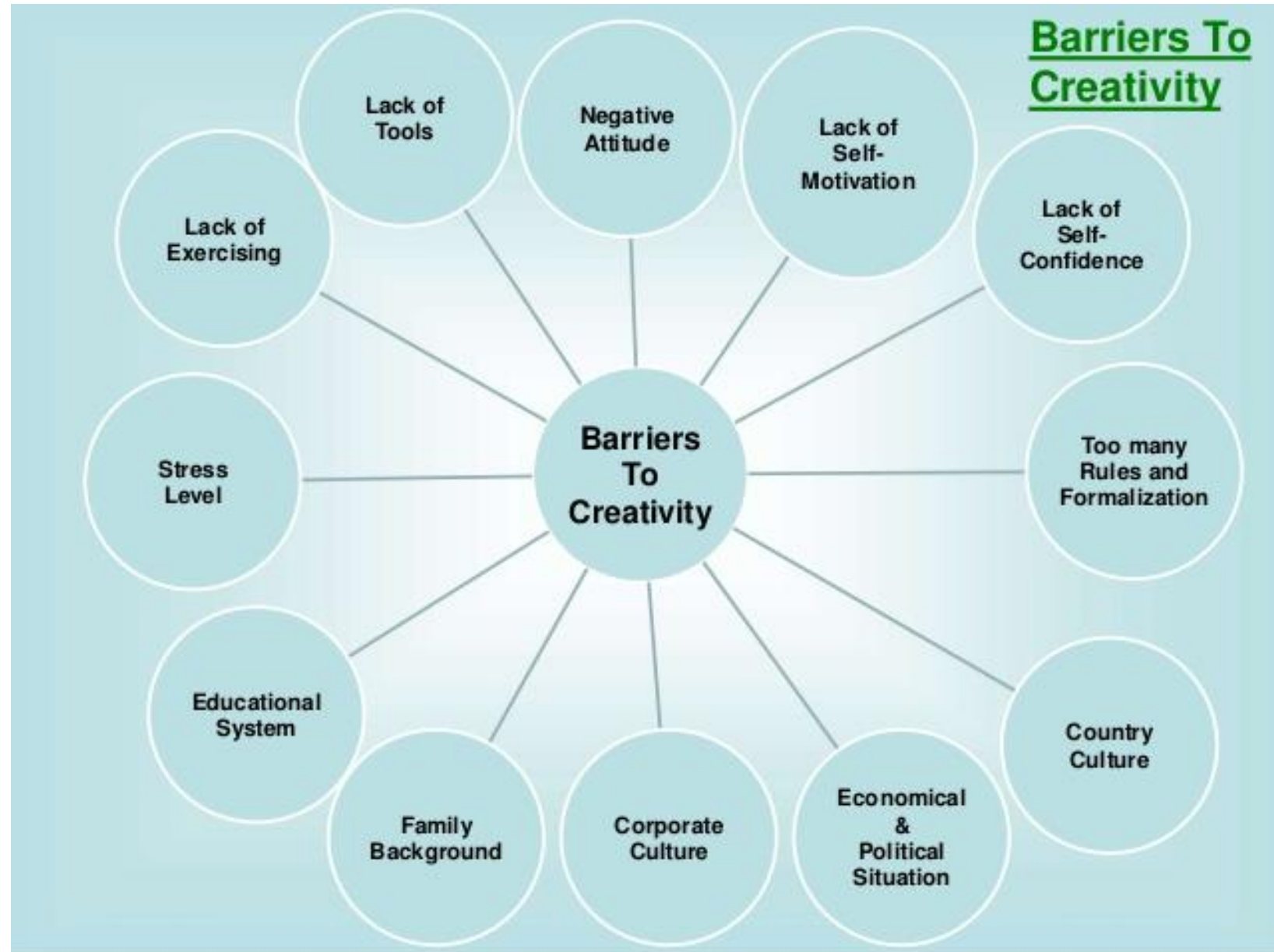


CREATIVITY AND PROBLEM SOLVING

- Creative thinkers are distinguished by their ability to synthesize new combinations of ideas and concepts into meaningful and useful forms.
- ✓ A creative engineer is one who produces a lot of ideas. These can be completely original ideas inspired by a discovery. More often, creative ideas result from putting existing ideas together in novel ways.
- ✓ A creative person is adept at breaking an idea down to take a fresh look at its parts, or in making connections between the current problem and seemingly unrelated observations or facts.
- ✓ We all like to be called “creative,” yet most of us, in our ignorance of the subject, feel that creativity is reserved for only the gifted few.
- ✓ There is the popular myth that creative ideas arrive with flash-like spontaneity—the flash of lightning and clap of thunder routine.

Aids to Creative Thinking

- *Develop a creative attitude*
- *Unlock your imagination*
- *Be persistent*
- *Develop an open mind*
- *Suspend your judgment*
- *Set problem boundaries*





Communication



WHAT IS COMMUNICATION ?

Communication is the process of sending and receiving messages through verbal or nonverbal means.



Articulate Design Ideas

At the surface this sounds easy – anyone can talk, but talking isn't the challenge. The challenge is in communicating well.

How can you communicate in a way that convinces your stakeholder(s) that this is the problem to focus on? The meaning behind a particular decision? The goal you're trying to achieve?

Knowing the goal is critical, but keep in mind that the goal can change.

As we progress through the design process, new insights arise which can influence the goal or direction of the project.

Articulate Design Ideas

A big part of any designers job is to **convincing others of our design can be key to success.**

Being a designer we need to consider consumers needs and values. This can be helpful in pinpointing what to focus on.

For example:

As an engineer, I value **technical constraints** and want to know the **technicality required** to make this project successful, so I should focus on how to address these.

As a product owner I value **business goals** and want to know how this is **solving the business problem**, so I should ensure that I outline how this solution will achieve that.

Equally important as understanding their needs and values is **listening**.

Listening helps to uncover the real problem and let others be heard.



Defining Designs' Purpose

1. **Solve a problem**

What problem does this particular design solve? Always refer to the problem, goals and big picture.

2. **Be easy for users**

How does this particular design affect the user? What barriers are we creating or minimizing?

3. **Be supported by everyone**

There will always be **someone** who will challenge your decision. Be prepared to explain why it's better than the alternative and the other directions you explored.



Documentation Benefits in Articulate Design

1. Avoids repeat conversations
2. Focuses on being articulate
3. Builds trust with stakeholders
4. Keeps the meeting on track



Those who tell the
stories rule society

Plato



Storytelling is a powerful communication tool

Evolution of Storytelling

Old - Less Immersive



New - More Immersive



PRINT



WORD OF MOUTH



MOBILE



VR / 360 / AR

Storytelling

Storytelling is used in design as a technique to get insight into users, build empathy and access them emotionally. With stories, designers speak via character, theme, melody, and spectacle to increase the appeal of what they offer and provide a solid understanding of the users.

Seven Elements in Storytelling

1. Plot – what are users trying to achieve/overcome?
2. Character – who are the users: not just demographically, but what insights do you require to understand what they're truly like?



[illegible]

- 23

Six Rules in Storytelling

1. Self knowledge and awareness
2. Clear structure and purpose
3. Have a character to root
4. Appeal to our deepest emotions
5. Surprising and unexpected
6. Simple and focused

SKETCHING

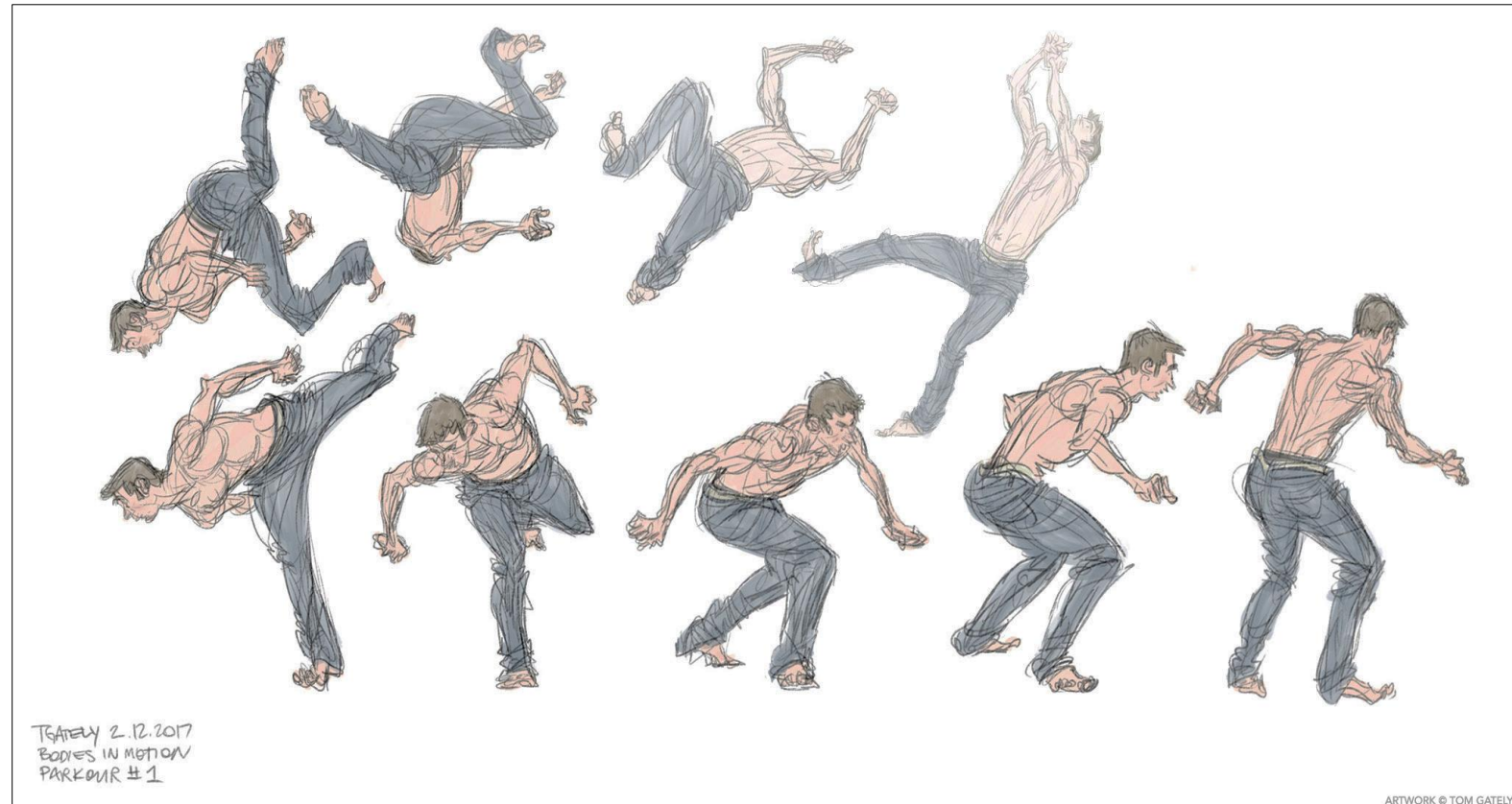
A sketch is a rapidly executed freehand drawing that is not usually intended as a finished work. A sketch may serve a number of purposes: it might record something that the artist sees, it might record or develop an idea for later use or it might be used as a quick way of graphically demonstrating an image, idea or principle.

Sketches can be made in any drawing medium. The term is most often applied to graphic work executed in a dry medium such as silverpoint, graphite, pencil, charcoal or pastel. It may also apply to drawings executed in pen and ink, digital input such as a digital pen, ballpoint pen, marker pen, water colour and oil paint. The latter two are generally referred to as "water colour sketches" and "oil sketches". A sculptor might model three-dimensional sketches in clay, plasticine or wax.





Dynamic drawings



How i DRAW DYNAMIC POSES!



① Know basic horse anatomy/proportions. ^{Q-key}



② THINK about the motion. (like a video!)



③ Gesture/silhouette
*don't worry about proportions/anatomy, just DRAW



④ Isolate Joints!
Anatomy is key!



⑤ Draw over!
Sketch (lol)



Jag 6201

K - Script

What is a k-script ?

The fastest and most flexible way to articulate a user experience

Blade Kotelly

UX Design Strategist, Consultant, Educator

Introduction

- In designing physical products, **sketching** is commonly used as a **quick** and **inexpensive** method to allow the designer to express an **idea quickly, evaluate the concept, and explain it** to others.
- Early sketches (which really are a form of prototyping) can be easily **tweaked, changed wholesale, edited, questioned or improved**.
- In the late 90s Mr. Kotelley realized that there was a need for UX designers to have a method to sketch — and he created a tool name “**K-Scripts**”.
- He started to apply this same idea to **web design, mobile application design, social-robot interaction design, and teaching** it in his design-thinking class at **MIT**. Since then, it has been used at numerous organizations including big ones like **Adobe and the U.S. Air Force**, for articulating interactive experiences for **software, hardware, robotics** and a whole lot more.

How To Make A K-Script

- K-Scripts are valuable when they focus on the 80% case — the most typical cases that users will encounter. They can also be used for **failure conditions** (in which a user may make mistakes, even if we don't expect to have it happen often) or any other context — but the best value is to show off a normal interaction and not a possible, but unlikely one.
- The format of the K-Script can be elegant and visually designed, or as simple as a spreadsheet table. Regardless of the format used, all K-Scripts contain **three essential elements**:

Who	Observable Action	Unobservable Action / Notes
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Who	Observable Action	Unobservable Action / Notes
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Who:

- The first section defines the actor, whomever is performing the action, and shows the sequential relationship between a user (or users) and the product. The who could be a generic person, a specific user, a product, a system or any entity that is involved in the interaction.

Observable Action:

- This section describes an action that a user performs or an action that occurs by a system — as long as it's something a viewer could observe. Whether it is discussing a physical interaction (touching a screen, walking down stairs) or a verbal one (talking to a machine), a detailed and truthful description matters here; the What section is what enables the reader to create a mental picture of the interaction.

Unobservable Action:

- This is the section that enables a designer to help the reader understand why a situation may be occurring (motivation of the user, or notes about how the system or technology works to enable this interaction.) It's also a good place for others to pose questions about the interaction.

Example: A Basic K-Script

Let’s take a look at a short K-Script for the design of a new ticketing kiosk at a large fair:

K-Scripts are an exceptionally easy thing to construct — and it would take a lot longer to sketch it out visually. The K-Script format also allows someone else to edit and provide feedback:

Who	Observable Action	Unobservable Action / Notes
Customer	Walks into the fair ground and sees a bunch of very modern touch-screen kiosks in a row, like at an airport. She sees the screen which has 2 buttons, “Buy Tickets”, “Make dinner reservations”	In research, we found that people generally want these 2 options, and we expect to have a limited number of staff there to handle other kinds of requests
Customer	Clicks on “Buy tickets”	
Kiosk	The screen shows “Number of Adults?” with a “+” and “-“ button and it’s default is set to “1”, and “Number of Children?” with a “+” and “-“ button and it’s default is set to “0” and a “Next” button	Prices aren’t displayed now, because there’s another screen which allows people to enter or scan discount codes/coupons
Customer	Hits the “Next” button	
Kiosk	Screen shows the price “\$24.00 Enter discount code, or scan a coupon” and a “Next” button	
Customer	Taps “Next”	
Kiosk	With the price still on the screen, directs the customer to swipe their credit card.	
Customer	Swipes the card	
Kiosk	Plays a success sound, shows a success message and a ticket is printed out	

When to use a K-Script ?



- The ability to K-Scripts up quickly is what makes them a powerful tool early on in the design process.
- A K-Script is a **great way to introduce a new idea**, as it can provide a clear picture of the interaction to those **who have no familiarity with the product**, but it can also be useful throughout the entire design process.
- Just as they function as an easy way to convey basic functionality, they can constantly be refined and become more robust (and more complete) over time, while still maintaining their flexibility.
- In addition to their use as a resource for the designer during product development, K-Scripts can also be used to **demonstrate functionality to clients or investors**.

When to use a K-Script ?

- By combining a thorough and well-developed K-Script with a rudimentary prototype, customers can **experience the feel of the system (through a functional prototype)** as well as understand the depth and complexity at which the **final system will** operate.
- And since **it's text and it's focused** on the **user experience**, anyone with a little understanding of the context will be able to understand it — **even non-designers**.
- Since K-Scripts are not meant to be all encompassing it's not necessary to create convoluted and overly complex scripts. The better use is to have the script focus on **typical interactions** and use more **detailed logic diagrams to show non-typical interactions**.
- However, additional K-Scripts should be created for less common or fringe cases when the script format is the best way to express the idea. A set of K-scripts creates a holistic view of the system that can communicate to lots of different groups and provide vision that **aligns teams**.